Owen Root

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Education

CUNY Graduate Center, New York, NY — *Expected May 2028*
Ph.D., Physics

- GPA: 3.64 / 4.00

Nebraska Wesleyan University, Lincoln, NE — *May 2023*
B.S., Physics; Minors in Mathematics and Political Science

- GPA: 3.62 / 4.00

Research Interests

Quantum information theory; quantum randomness and foundations; event-based / neuromorphic vision; computational optics; machine learning for imaging; numerical methods for quantum dynamics.

Publications

Jessica Burns, **Owen Root**, Hui Jing, and Imran M. Mirza, "Engineering optomechanically induced transparency by coupling a qubit to a spinning resonator," *J. Opt. Soc. Am. B* **40**, 958–965 (2023).

Preprints and Other Publications

Owen Root and Maria Becker, "Does true randomness exist? Efficacy testing IBM quantum computers via statistical randomness," arXiv:2401.12250 (2024).

Owen Root, "Expression for $g(k)$ related to Waring's problem," arXiv:2508.17950 (2025).

Owen Root, "Robocode battle builds bridges," *The SPS Observer* **56**(3), Society of Physics Students / American Institute of Physics, 1 Feb. 2023. Available online: https://students.aip.org/observer/robocode-battle-builds-bridges

Research Experience

Graduate Research Assistant — *New York, NY*
CUNY Graduate Center and Hunter College — *Aug 2023 – Present*

- Developed an analytical statistical model for neuromorphic / event-based cameras that jointly describes signal-driven and noise-induced events.
- Designed experimental methods to infer unknown physical properties of an event camera from recorded event streams using the analytical model.
- Trained U-Net architectures with self attention to reconstruct still images from event-camera noise for model validation.
- Developed a quantum algorithm that distinguishes phase-modified unitary oracles using a single-qubit measurement.
- Built a numerical simulation suite for time-dependent Schrödinger dynamics with time-dependent and non-trivial boundary conditions.
- Implemented analysis tools for numerical simulations of quantum dynamics in large biological molecules.
- Created a program that uses genetic programming to infer analytic functional forms from hand-drawn plots.

Undergraduate Research Assistant — *Lincoln, NE*
Nebraska Wesleyan University — *Aug 2020 – May 2023*

- Investigating statistical randomness in IBM quantum computers:
  - Used IBM quantum devices to generate random bit strings from measurements of superposition states.

- Applied statistical tests grounded in probability theory to quantify deviations from ideal randomness and to compare with classical pseudo-random number generators.
- Discussed implications for cryptography and other applications that require high-quality randomness from quantum devices.

- Simulation and investigation of quantum phenomena using an acoustic double slit:
  - Designed and operated an acoustic double-slit apparatus with controlled phase and timing of sound waves.
  - Used acoustic interference patterns as an accessible analog to illustrate the wave nature of quantum mechanics.
  - Examined how loss of coherence suppresses interference and connected these effects to limitations on quantum technologies such as quantum computers.

Physics REU Student *Oxford, OH*
Miami University *July 2022 – Aug 2022*

- Investigated probe-light transmission in a spinning optomechanical ring resonator coupled to a single qubit with applications to nonreciprocal photonic devices and quantum information processing.
- Analyzed how strong qubit–resonator coupling modifies transmission for clockwise versus counterclockwise rotation.
- Demonstrated that a strongly coupled qubit enhances transmission only for clockwise rotation, indicating a direction-dependent optical response.

Awards and Fellowships

Graduate Assistant Fellowship for graduate study, CUNY Graduate Center and Hunter College — Aug 2023 – Present

French Foundation Scholarship for collegiate study — Aug 2019 – Present

STEM Scholarship for undergraduate study, Nebraska Wesleyan University — Aug 2019 – May 2023

Conferences

Nebraska Research & Innovation Conference — 2024
- Contributed poster presentation.

Optica Frontiers in Optics — 2022
- Contributed oral presentation.

Society of Physics Students Physics and Astronomy Congress — 2022
- Contributed poster presentation.

Leadership and Service

President, Society of Physics Students (SPS) Chapter, Nebraska Wesleyan University — May 2021 – May 2023
- Led revitalization of the local chapter following the COVID-19 pandemic.
- Organized multiple interdepartmental student events and recruitment activities.
- Planned and executed a chapter trip to the 2022 SPS Physics and Astronomy Congress in Washington, DC.

Teaching Experience

Graduate Adjunct Lecturer, Department of Physics and Astronomy, Hunter College — Aug 2024 – Present
- Prepare and lead weekly instructional laboratories for introductory physics courses.
- Grade lab reports and practical exams and collaborate with course instructors on final grades.

Physics Tutor, Cooper Center, Nebraska Wesleyan University — Aug 2021 – May 2023
- Provided one-on-one and small-group tutoring for undergraduate physics students.
- Developed study strategies and review sessions to help students prepare for exams.

Skills

Programming and data analysis: Python, Qiskit, LaTeX, Mathematica, R.

Operating systems: Windows, macOS, UNIX / Linux.

Media software: Adobe Photoshop, Adobe Premiere.